

Zakhar Pashkin

Senior Computer Vision Engineer | AI Product Engineer | OCR, Segmentation, Detection, VLM/LLM Systems

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SUMMARY

Senior Computer Vision Engineer and AI Product Engineer with 7+ years shipping OCR, segmentation, detection, multimodal search, VLM/LLM workflows, backend services, and production ML systems. Strongest where model quality, product behavior, operator workflow, and launch reliability need one owner: dataset quality, model selection, API delivery, UI surfaces, monitoring, review gates, leak checks, and rollback-ready releases.

CORE STACK

AI products: VLM/LLM workflows, OpenAI APIs, agent automation, Telegram mini apps, Chrome extensions, review gates

Modeling: PyTorch, TensorFlow/Keras, OpenCV, YOLO, MMDetection, CLIP/VLMs, Transformers, ONNX conversion

Production: FastAPI, Tornado, REST/gRPC, Docker, Kubernetes, GitHub Actions, GCP, AWS, Cloud Run, monitoring, CI/CD, MLOps

Computer vision: OCR, segmentation, detection, landmarking, tracking, face texture analysis, neural search, multimodal CV

Edge/mobile: ONNX, TFLite, CoreML, TensorRT, PyTorch Mobile, quantization, mobile inference profiling, Flutter/Android/iOS integration

EXPERIENCE

Senior Computer Vision Engineer & AI Product Engineer

Jun 2022 - Present

Various Startups - selected startup, independent, and direct client engagements

- Delivered AI product systems across automation, segmentation, detection, OCR, landmarking, face search, neural search, multimodal video search, VLM/LLM workflows, data pipelines, and edge deployment.
- Built or modernized models, APIs, operator workflows, and user-facing product surfaces where scale, latency, and release reliability mattered.
- Optimized PyTorch/TensorFlow models for ONNX, TFLite, and CoreML deployment, with mobile inference targets up to 50 to 60 FPS in OCR and segmentation flows.
- Reduced labeling cost by up to 80% with active-learning and auto-annotation pipelines; released tooling around interactive segmentation and OCR workflows.
- Delivered FastAPI/Tornado services, Docker/Kubernetes deployments, GCP/AWS infrastructure, CI/CD, and monitoring for production ML, AI automation, and launch workflows.
- Shipped public OpenClaw/ClawHub release systems, including a downloads tracker showing 3,745 public ClawHub downloads across 11 packages as of 2026-05-14.
- Added review gates for high-risk AI delivery: benchmark checks, human approvals, browser evidence, public-surface leak checks, rollback criteria, and promotion decisions before release.

Computer Vision Engineer

Jun 2019 - Jun 2022

CFT - Akademgorodok, Novosibirsk

- Developed classification, object detection, OCR, segmentation, and document-recognition algorithms for fintech and mobile products.
- Optimized energy-meter OCR and segmentation models for stable 50 to 60 FPS on mid-range smartphones.
- Built masked face anti-spoofing and Cyrillic offline signature-verification research systems with deployment-focused error analysis.
- Improved specialized mobile-GPU model quality with neural architecture search and deployment-focused tuning.
- Implemented CLIP pre-validation and Toloka automation to reduce labeling cost and review load through click-driven segmentation workflows.

Computer Vision Project Mentor

Nov 2019 - Jun 2022

CFT apprenticeship corporate program - Part-time

- Mentored pre-internship engineers on production-style CV projects including mobile energy-meter recognition, crowdsourcing with neural networks, and Cyrillic handwriting OCR.
- Reviewed model choices, dataset quality, error analysis, and deployment constraints so student projects could pass engineering review rather than remain demos.

Machine Learning Project Mentor

Oct 2018 - Oct 2019

School of AI - Remote / San Francisco Bay Area community

- Built and supported ML/DL software for analytics and online education, including Flutter product work and GCP infrastructure.
- Helped secure and operate Google Cloud support for a global AI education platform while coordinating remote contributors and volunteers.

SELECTED PROJECT EVIDENCE

Fast OCR ONNX Inference Server

Three-stage OCR API: line segmentation, word segmentation, CRNN text recognition, Docker packaging, and JSON boxes/text responses.

Multimodal Video Search Platform

Keyframes, ASR/OCR, objects, faces, visual embeddings, transcript embeddings, Qdrant/Postgres retrieval, and quality-agent checks.

Full-Face Wrinkle Segmentation Lab

Cosmetic face-region masks, YOLO segmentation, skeletonized line traces, overlays, CSV outputs, timing events, and visual QA gates.

Dermaself Flutter Skin Analysis App

Flutter/Firebase guided capture app with onboarding, questionnaire intake, offline model runtime, ROI gates, and results screens.

Nutrition Label OCR Mobile Lab

SwiftUI food detection and nutrition-label OCR using ML Kit, crop-assisted text recognition, table parsing, and structured results.

CV Repro Lab Skills

Reproducible CV experiment workflow with benchmark gates, review dashboards, browser validation, and 913 public ClawHub downloads across two packages.

RECRUITER EVIDENCE CHECKS

AI product scope: Owns model, backend, workflow, UI, and launch reliability rather than stopping at notebooks or demos.

CV depth: Can explain data, metrics, failure modes, conversion path, and edge/runtime constraints for shipped OCR, segmentation, and detection work.

Evidence quality: Public work links to repos, releases, dated counters, and review gates; non-public CV work is summarized without repo or live-service links.

EDUCATION & CERTIFICATIONS

Siberian Federal University, 2007-2012. Additional professional training includes TensorFlow, Deep Learning and Computer Vision, SQL, JavaScript, Kubernetes on Google Cloud, and PyTorch scholarship work.